

Data Handling and Infrastructure for AI

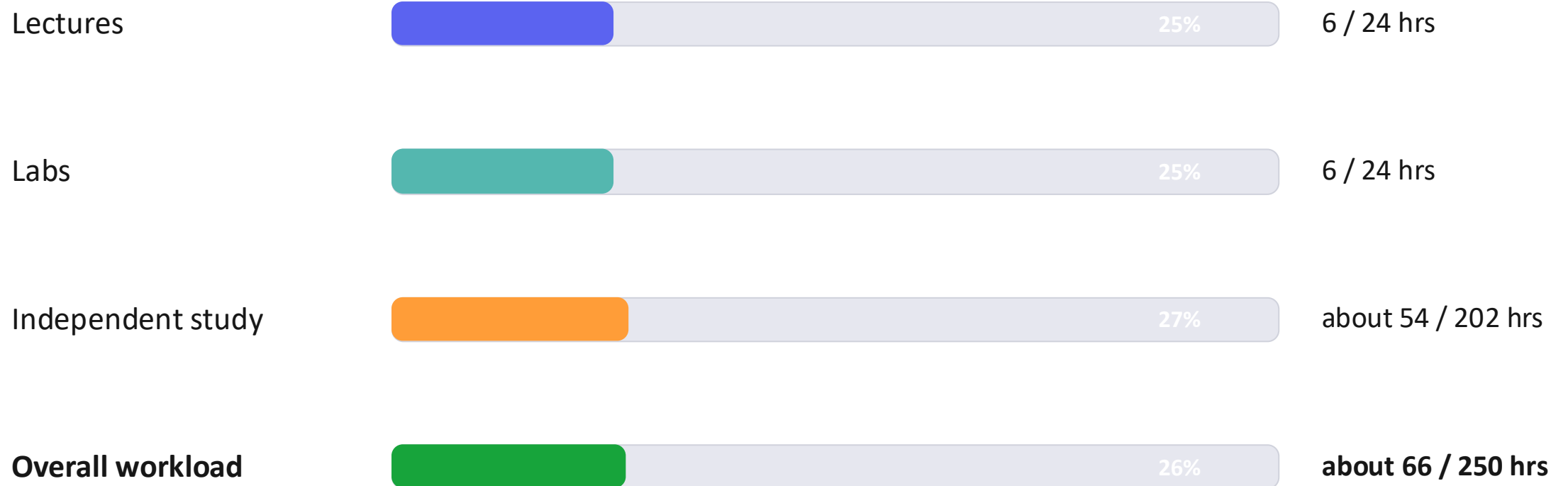
Statistical Reasoning and Exploratory Data Analysis

Lecture 4

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SETU | 2026

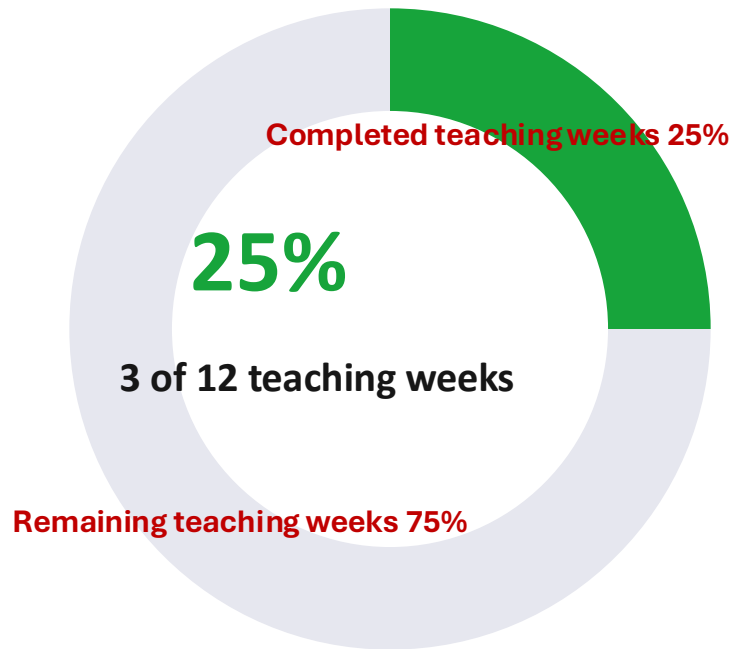
Workload and Progress at the Start of Week 4



Reading week adds project research and independent study, but no new contact hours.

Approximately 26% of the expected module workload has elapsed

Course Coverage So Far



Week 1

Data types

Jupyter notebooks

Week 2

Data storage

Pandas introduction

Week 3

Vector DBs and data lakes

GCS and database lab

Reading week checkpoint: select the dataset, define the task and justify the initial storage design

Week 4 in the Course Plan



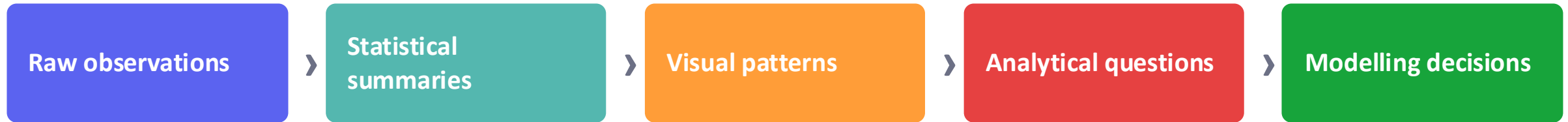
Statistics explains what a summary means; pandas makes the analysis repeatable

Learning Objectives

- Define an estimand and distinguish parameters, estimators and realised estimates.
- Explain sampling distributions, standard errors, confidence intervals and bootstrap uncertainty.
- Use effect sizes, robust summaries and conditional analysis instead of relying on p-values alone.
- Recognise multiplicity, dependence, missing-data assumptions, distribution shift and leakage.
- Build concise, vectorised and reproducible Pandas analyses for model-oriented EDA.

The emphasis is statistical judgement for machine-learning data, not formula recall

Why Statistical Foundations Come Before EDA



Statistics helps us

- Compress many observations
- Quantify uncertainty
- Compare groups

Statistics cannot decide

- Whether data is representative
- Whether an association is causal
- Whether a variable is ethically appropriate

Population, Sample and Observation

Population

All cases about which we want to draw a conclusion

Sample

The observed subset available for analysis

Observation

One measured unit, such as one passenger

Titanic example: the 891 recorded passengers form the sample. The population depends on the question we ask.

A precise research question defines the target population

Variable Types Guide Statistical Choices

Numerical continuous

Age, Fare

Means, medians, spread, histograms

Numerical discrete

SibSp, Parch

Counts, proportions, grouped summaries

Categorical nominal

Sex, Embarked

Frequencies, proportions, cross-tabs

Categorical ordinal

Pclass

Ordered comparisons; spacing is not necessarily equal

Identifier

PassengerId, Ticket

Linking and grouping, usually not direct modelling inputs

Data type describes meaning, not merely the pandas dtype

Descriptive statistics

- Summarise the observed data
- Describe centre, spread and shape
- Compare recorded groups
- Do not extend beyond the sample

Inferential statistics

- Use a sample to reason about a population
- Quantify sampling uncertainty
- Test or estimate population patterns
- Depend on assumptions and sampling design

Measures of Centre

Example fares

10

12

12

13

15

60

Mean = 20.3

Median = 12.5

Mode = 12

The value 60 pulls the mean upward. The median better represents the typical value in this skewed example.

Choose a summary that matches the distribution and analytical question

Mean, Median and Mode in Pandas

```
age_mean = df['Age'].mean()
age_median = df['Age'].median()
age_mode = df['Age'].mode().iloc[0]

print(f'Mean: {age_mean:.2f}')
print(f'Median: {age_median:.2f}')
print(f'Mode: {age_mode:.2f}')
```

Titanic Age is approximately: mean 29.70 years, median 28 years, mode 24 years.

pandas excludes missing values by default in these calculations

Measures of Spread

Range

maximum minus minimum

Simple but controlled by extremes

Interquartile range

Q3 minus Q1

Spread of the middle 50%

Variance

average squared deviation from the mean

Useful mathematically; squared units

Standard deviation

square root of variance

Typical distance from the mean when interpretation is appropriate

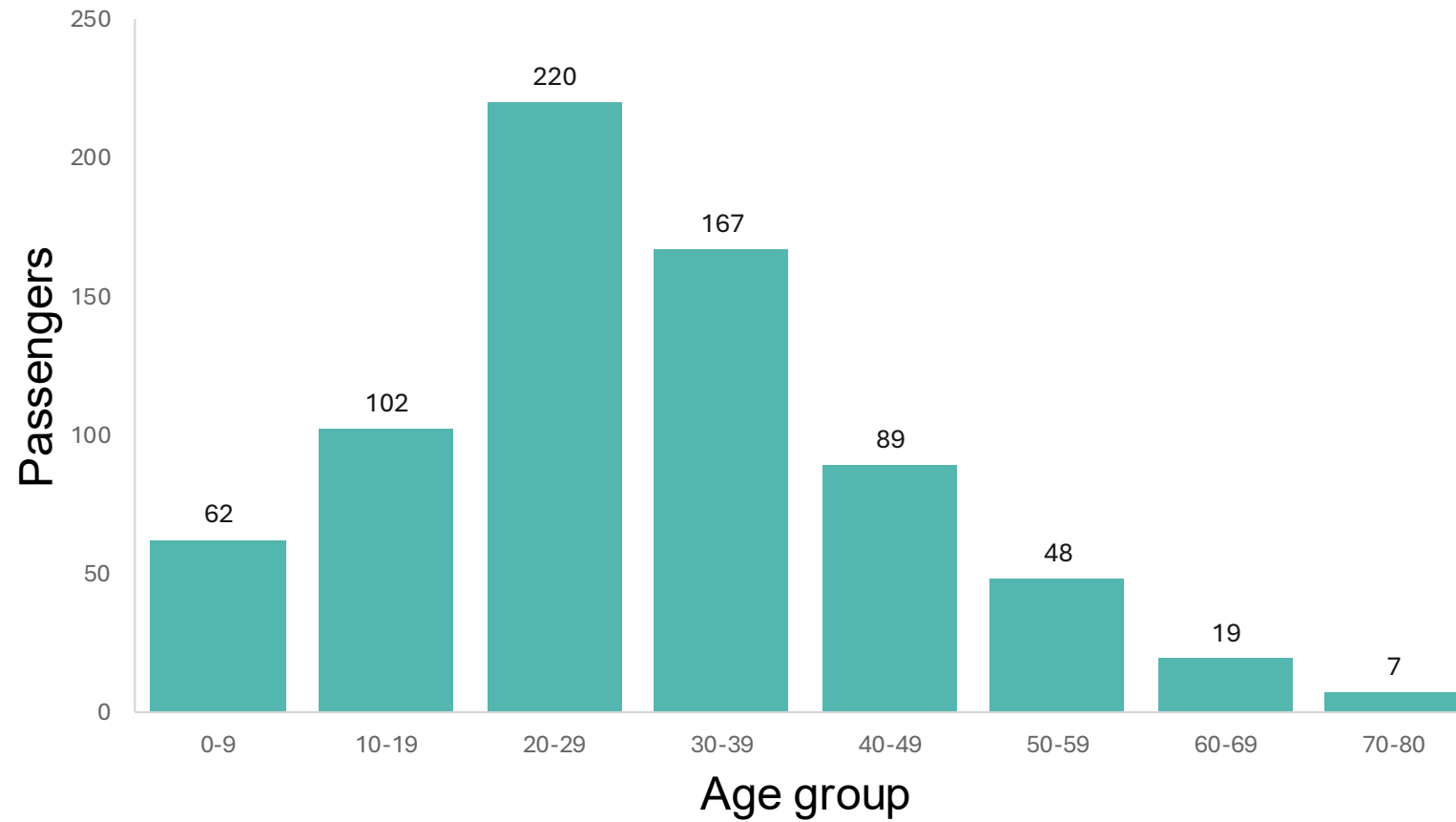
Centre without spread can hide major differences between datasets

```
age = df['Age'].dropna()

age_range = age.max() - age.min()
q1 = age.quantile(0.25)
q3 = age.quantile(0.75)
iqr = q3 - q1
variance = age.var()      # sample variance
std_dev = age.std()       # sample standard deviation
```

Always report the unit and whether the calculation describes a sample or a full population

Distribution of Passenger Age



Shape

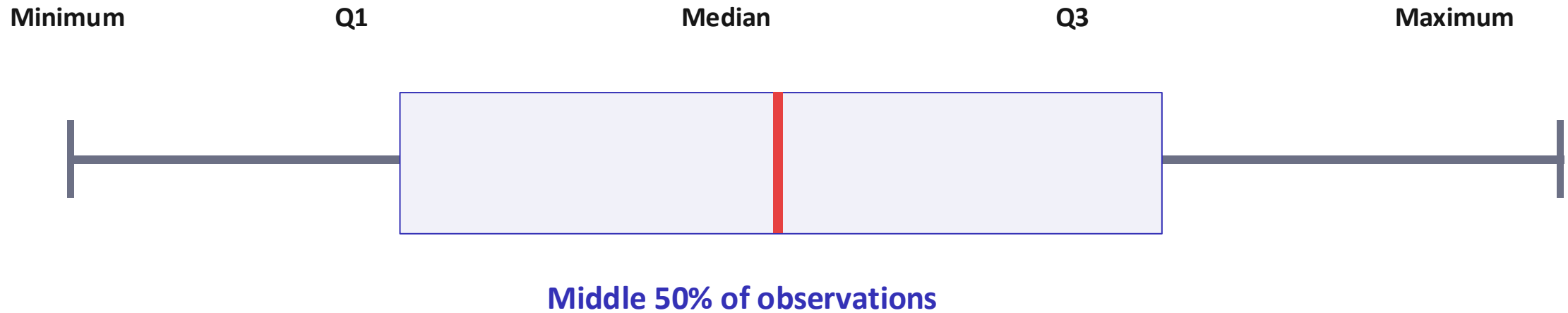
Most recorded ages lie between 20 and 39.

Missingness

177 of 891 Age values are missing.

A histogram shows frequency across intervals; bin choices affect the visual pattern

Quartiles and the Boxplot



Potential outliers are commonly flagged beyond $1.5 \times \text{IQR}$, but domain context determines whether they are errors.

A boxplot summarises position and spread; it does not show every feature of the distribution

Symmetric distribution

- Mean and median are similar
- Both tails have comparable shape
- Standard deviation may be informative

Right-skewed distribution

- A long upper tail raises the mean
- Median and IQR are often more representative
- Titanic Fare is strongly right-skewed

Skewness describes asymmetry. It does not by itself imply that the data is invalid.

Match the summary statistic to the observed shape

$$z = (x - \text{mean}) / \text{standard deviation}$$

Interpretation

- $z = 0$: at the mean
- $z = 1$: one standard deviation above
- $z = -2$: two standard deviations below

Cautions

- Sensitive to extreme observations
- Less intuitive for strongly skewed variables
- A large absolute z score is a flag, not proof of error

Standardisation places variables on a comparable scale but does not make them normally distributed

Sensitive and Resistant Summaries

Statistic	Affected strongly by extremes?	Useful when
Mean	Yes	Distribution is reasonably symmetric
Median	No	Distribution is skewed or contains valid extremes
Standard deviation	Yes	Mean based variation is meaningful
IQR	No	A resistant measure of spread is needed

Use several summaries when one number cannot describe the distribution adequately

```
counts = df['Embarked'].value_counts(dropna=False)
proportions =
df['Embarked'].value_counts(normalize=True,
dropna=False)

summary = pd.DataFrame({
    'count': counts,
    'proportion': proportions
})
```

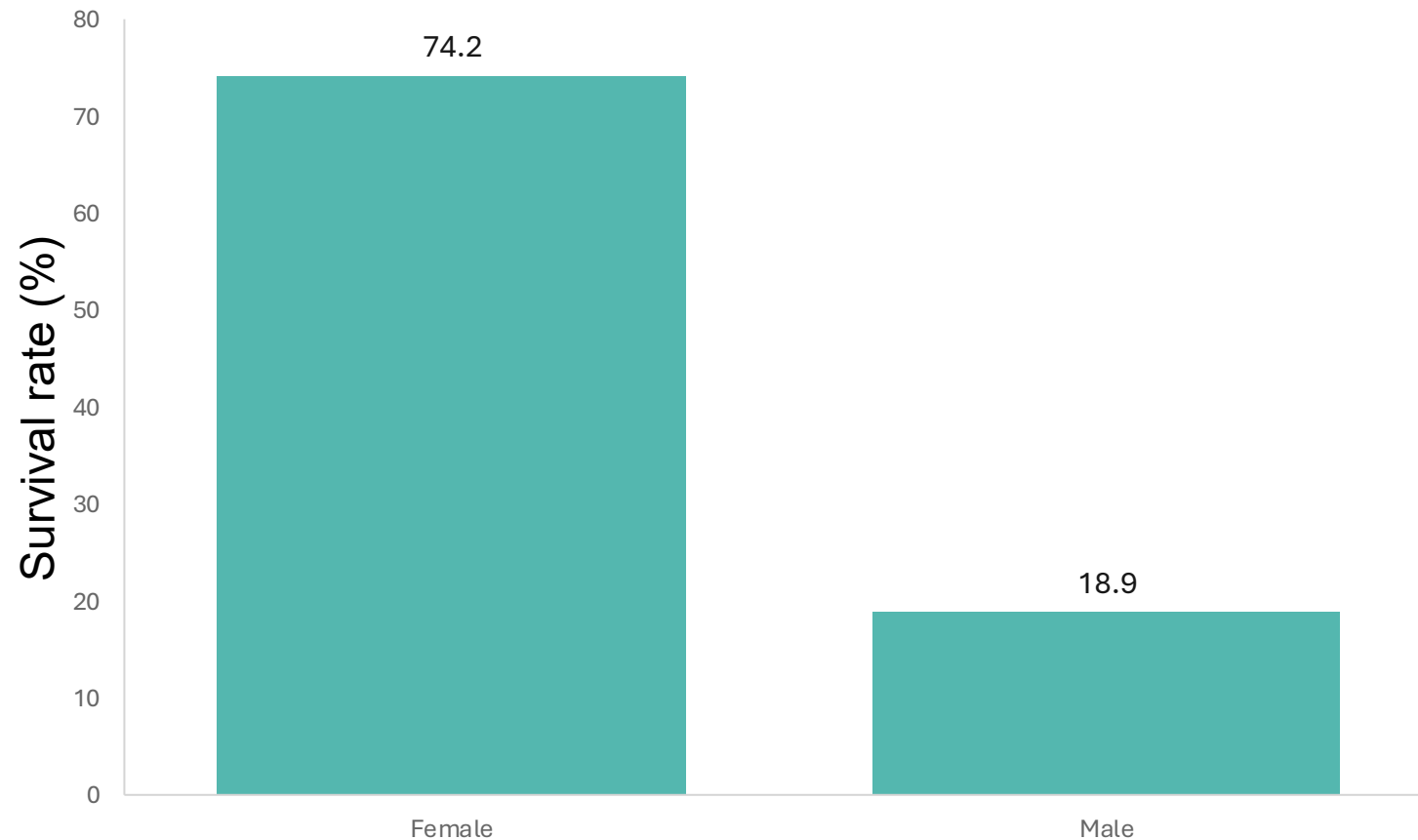
Counts answer: how many?

Proportions answer: what share?

Keep missing values visible during initial profiling.

A frequency table is the categorical equivalent of a numerical summary

Cross Tabulation: Survival by Sex

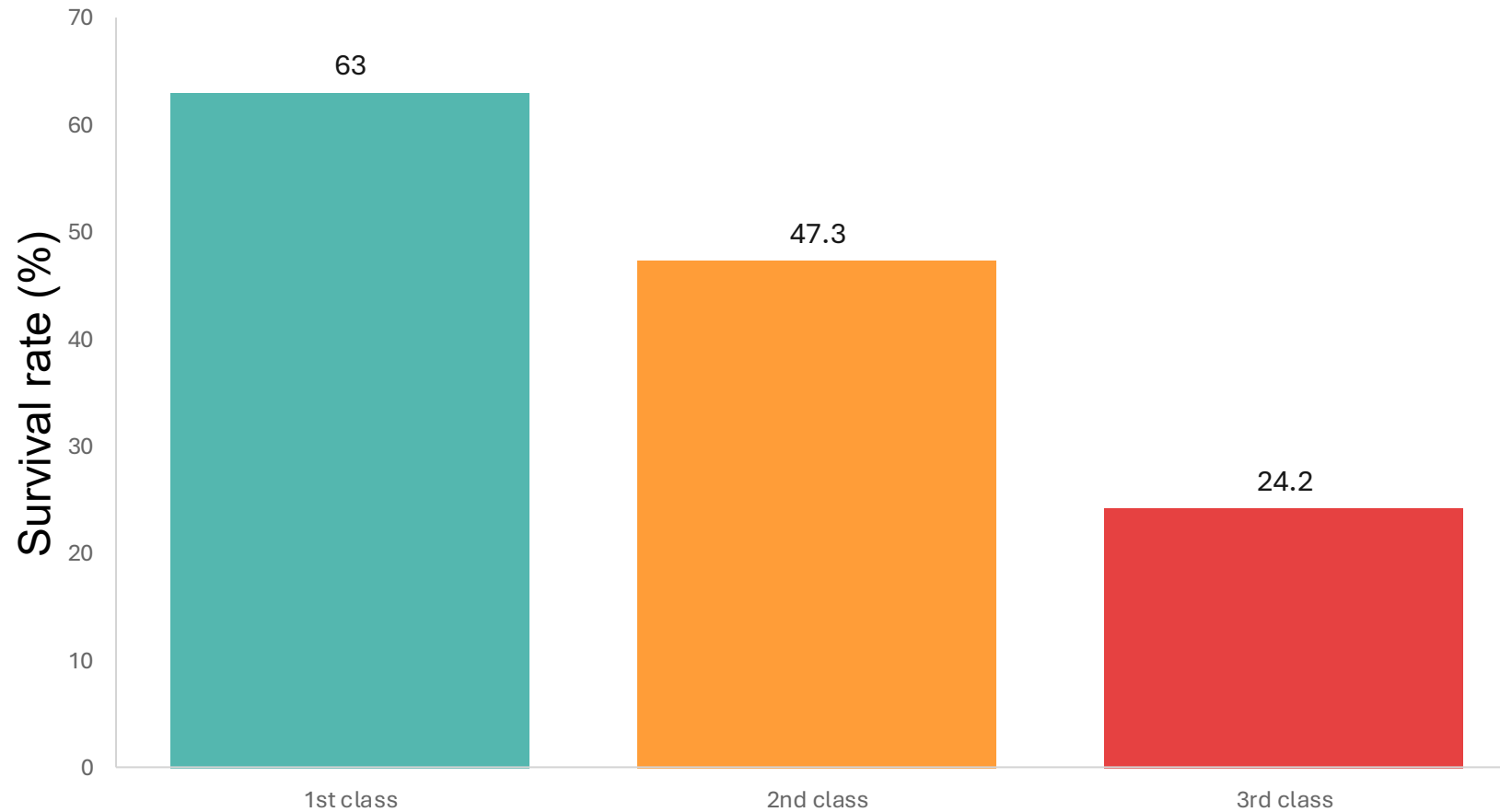


```
pd.crosstab(  
    df['Sex'],  
    df['Survived'],  
    normalize='index'  
) * 100
```

Observed association, not a causal estimate

Compare proportions within each group, not only raw counts

Grouped Comparison: Survival by Passenger Class

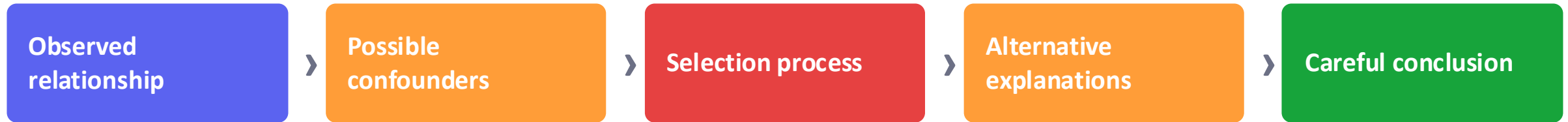


Questions to ask

- Are group sizes similar?
- Could age or sex explain part of the pattern?
- Does class represent several underlying factors?

Group comparisons motivate further analysis; they do not isolate causal effects

Association Does Not Establish Causation



Example: passenger class is associated with survival, but class also relates to cabin location, access to lifeboats, age and family composition.

Describe what the data shows and separate it from the explanation you hypothesise

Correlation measures the direction and strength of a linear relationship.

Pearson correlation

- Measures linear association
- Sensitive to outliers
- Requires numerical variables

Spearman correlation

- Uses ranks rather than raw values
- Captures monotonic relationships
- Often more resistant to extremes

```
df[['Age', 'Fare', 'SibSp', 'Parch']].corr(method='pearson')
```

Correlation near zero does not rule out a non-linear relationship

Potential sources

- Coverage: some groups cannot enter the sample
- Selection: inclusion depends on behaviour or access
- Survivorship: only remaining cases are observed
- Measurement: recording quality differs by group

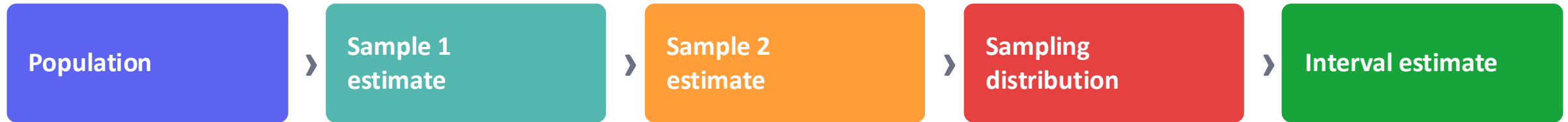
Questions for the project

- Who generated the data?
- Who or what is absent?
- What period and setting does it represent?
- How does deployment differ from collection?

A large sample can estimate the wrong population very precisely

Sampling Variability and Confidence Intervals

Different random samples from the same population produce different estimates.



A confidence interval expresses uncertainty in an estimation procedure. It does not guarantee that the population parameter lies in one realised interval.

Report uncertainty when a sample supports conclusions about a wider population

Hypothesis Testing: A Careful Introduction

Question

- Could an observed difference arise under a stated null model?
- What assumptions define the reference distribution?
- How large is the effect, not only whether $p < 0.05$?

Avoid

- Treating a p-value as the probability that the null is true
- Equating statistical significance with practical importance
- Testing many relationships without adjustment or rationale

EDA generates questions; confirmatory testing requires a pre-specified design

Estimand, Estimator and Estimate

Estimand

The population quantity the question asks for

Difference in survival probability between passenger classes

Estimator

The rule applied to a sample

Difference between two sample proportions

Estimate

The realised numerical result

0.388 in this dataset

Start with the estimand: a precise question determines the statistic, comparison and uncertainty

Mean squared error = bias² + variance + irreducible noise

Bias

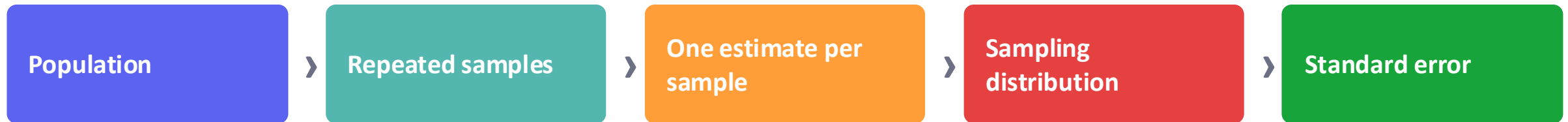
- Systematic error across repeated samples
- Can arise from selection, measurement or modelling assumptions
- More data does not remove design bias

Variance

- How much an estimator changes across samples
- Usually decreases as sample size grows
- Regularisation can trade some bias for lower variance

An estimate can be precise but biased, or unbiased but too variable to be useful

Sampling Distributions and Standard Error



The standard deviation describes variation among observations.

The standard error describes variation of an estimator across samples.

```
# standard error of a sample mean  
se_age = df['Age'].std() / np.sqrt(df['Age'].count())
```

Do not use standard deviation and standard error interchangeably

Law of large numbers

- The sample mean stabilises around the expected value as n increases
- Requires observations compatible with the sampling assumptions
- Does not repair biased collection

Central limit theorem

- Many standardised estimators approach a normal sampling distribution
- Supports approximate intervals and tests
- Convergence can be poor for heavy tails, dependence or small n

Large n improves sampling precision; it does not guarantee validity or practical importance

Bootstrap Confidence Intervals



```
rng = np.random.default_rng(42)
rates = [
    df.sample(len(df), replace=True, random_state=int(rng.integers(1e9)))
    .groupby('Sex')['Survived'].mean().diff().iloc[-1]
    for _ in range(2000)
]
ci = np.quantile(rates, [0.025, 0.975])
```

Bootstrap the sampling unit—not individual rows—when observations are clustered

Effect Size Before Statistical Significance

Difference in means

Outcome units

How far apart are two group averages?

Standardised mean difference

Standard deviation units

How large is the shift relative to within-group spread?

Risk difference

Percentage points

How much does event probability change?

Risk ratio / odds ratio

Relative scale

How many times larger is risk or odds?

Report magnitude and uncertainty; a tiny effect can be statistically significant in a large dataset

If 20 independent null hypotheses are tested at $\alpha = 0.05$, at least one false positive is likely about 64% of the time.

Control the search

- Pre-specify primary questions
- Limit interaction hunting
- Separate exploration from confirmation

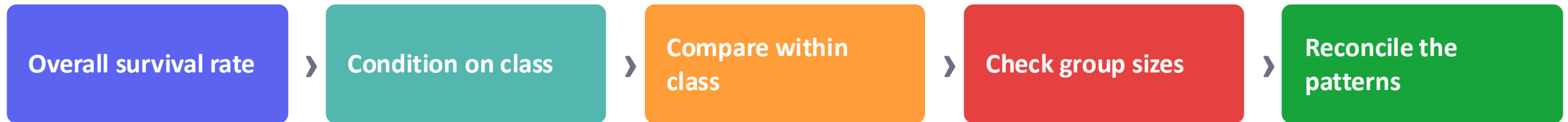
Control error rates

- Bonferroni controls family-wise error
- Benjamini-Hochberg controls false discovery rate
- Validate selected findings on held-out data

EDA may generate hypotheses; it should not disguise them as pre-specified confirmation

Conditional Analysis and Simpson's Paradox

An aggregate relationship can weaken, disappear or reverse after conditioning on a third variable.



Why reversal happens

- Groups have different compositions
- The conditioning variable relates to exposure and outcome

Pandas pattern

- `groupby(['Pclass','Sex']).agg(...)`
- Use counts beside rates and intervals

Do not interpret a marginal statistic when the deployment decision is conditional

Missingness as a Statistical Problem

MCAR

Missingness unrelated to observed and unobserved values

Complete-case estimates may remain unbiased but lose precision

MAR

Missingness explained by observed variables

Model or impute conditionally on observed information

MNAR

Missingness depends on unobserved values or causes

Sensitivity analysis and domain evidence are essential

A missingness indicator can reveal structure, but it does not identify the missing-data mechanism

Dependence

Rows share person, household, time, site or source

Use grouped summaries, clustered uncertainty and group-aware splits

Distribution shift

Training and deployment distributions differ

Compare prevalence, quantiles, categories and missingness by split or period

Leakage

Features or transformations use unavailable information

Define feature-availability time and fit preprocessing on training data only

The independent and identically distributed assumption is an engineering claim about how the data was generated

Probability as the Language of Uncertainty

Marginal probability

$P(\text{Survived})$

Probability of survival for a randomly selected recorded passenger

Joint probability

$P(\text{Survived and Pclass}=1)$

Probability that both events occur

Conditional probability

$P(\text{Survived} \mid \text{Pclass}=1)$

Probability of survival within first class

Complement

$1 - P(\text{Survived})$

Probability of the event not occurring

Every reported proportion should identify its conditioning set and denominator

$$P(A \mid B) = P(B \mid A) P(A) / P(B)$$

Why it matters for AI

- Converts likelihoods into posterior probabilities
- Shows how class prevalence affects interpretation
- Explains why predictive values change across populations

Diagnostic example

- Sensitivity is not the same as $P(\text{condition} \mid \text{positive})$
- A rare condition can yield many false positives
- Deployment prevalence must be considered

Model scores need the deployment base rate before they can support calibrated decisions

Power = P(rejecting the null when a meaningful effect is present)

Effect size

Smaller effects require more information

Sample size

Larger effective n usually increases power

Noise

Greater variability reduces signal-to-noise ratio

Significance threshold

Stricter thresholds reduce false positives but also power

Dependence

Correlated rows contribute less information than independent rows

Plan for the smallest effect that would change a decision, not merely for $p < 0.05$

Chi-Square Test: What Question Does It Answer?

Research question

Are passenger class and survival statistically independent?

H_0

Passenger class and survival are independent in the population represented by the sampling design.

Evidence

Compare observed cell counts with the counts expected if independence were true.

Output

Report χ^2 , degrees of freedom, expected counts, p-value and Cramér's V.

The test concerns association between categorical variables; it does not estimate a causal effect

Expected Counts Under Independence

Passenger class	Did not survive	Survived	Row total
First	80	136	216
Second	97	87	184
Third	372	119	491
Column total	549	342	891

Expected count for Third class and Survived

$$E = (491 \times 342) / 891 \approx 188.5$$

Observed = 119

Expected $\approx$ 188.5

This cell contributes substantially to χ^2 because the observed and expected counts differ.

Expected counts describe the independence model; they are not predictions for individual passengers

From Observed Differences to the Chi-Square Statistic

$$\chi^2 = \sum (O - E)^2 / E$$

- Each cell contributes a non-negative quantity.
- Larger observed–expected discrepancies increase χ^2 .
- The division by E scales the discrepancy relative to the expected count.
- Sparse expected cells can make the approximation unreliable.

Assumption check

Check	Interpretation
Independent observations	Each unit contributes once
Expected counts	Prefer ≥ 5 in most cells
Sparse table	Consider Fisher's exact test

The chi-square statistic measures departure from independence, not practical importance

p-Value and Cramér's V Answer Different Questions

p-value

How incompatible are the data with the independence model, given the assumptions?

Affected strongly by sample size

Cramér's V

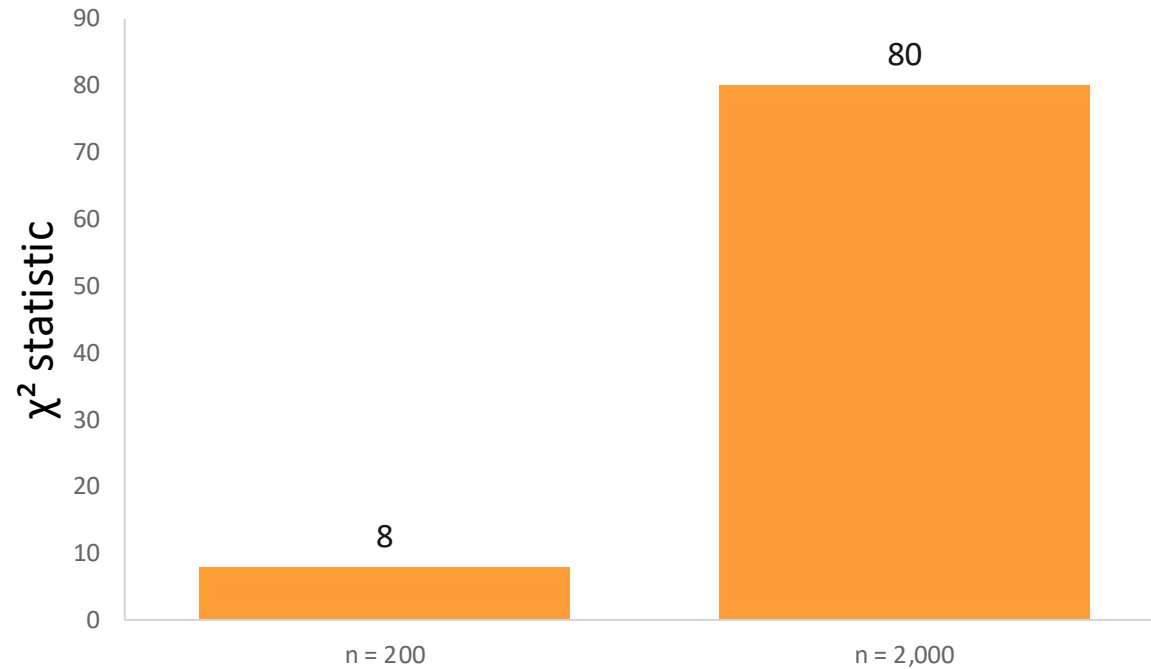
How large is the categorical association relative to the table dimensions?

$$V = \sqrt{\chi^2 / (n \times \min(r-1, c-1))}$$

Approximate V	Context-dependent description
0.10	weak
0.30	moderate
0.50	strong

Report the contingency table, p-value and effect size together

Why Sample Size Changes the p-Value



Same proportions; ten times the counts

Same association strength

Quantity	n = 200	n = 2,000
Cramér's V	0.20	0.20
χ^2	8	80
p-value	≈ 0.0047	$< 10^{-18}$

- Larger samples detect smaller departures from independence.
- A smaller p-value does not imply a stronger association.
- Observational association can reflect confounding or selection.

For Titanic: describe the association; do not claim that passenger class itself caused survival

Choose a Test by Question and Assumptions

Two numerical groups

Welch t-test

Independent observations; interprets a mean difference

Skewed or ordinal groups

Mann-Whitney U

Compares rank distributions; not automatically a median test

Paired measurements

Paired t-test / Wilcoxon

Pairs must represent the same unit or matched units

Two categorical variables

Chi-square / Fisher's exact

Expected cell counts and sampling design matter

Several groups

ANOVA / Kruskal-Wallis

Follow-up comparisons require multiplicity control

The test follows the estimand, design and assumptions; it does not define the research question

Chi-square test

- Compares observed and expected cell counts
- Tests independence under a reference model
- Sensitive to sample size and sparse cells

Effect size

- Phi for a 2×2 table
- Cramer's V for larger contingency tables
- Always inspect the underlying proportions and counts

```
table = pd.crosstab(df['Pclass'], df['Survived'])  
chi2, p, dof, expected = scipy.stats.chi2_contingency(table)  
cramers_v = np.sqrt(chi2 / (table.values.sum() * min(table.shape[0]-1, table.shape[1]-1)))
```

A small p-value indicates incompatibility with independence, not the strength or cause of association

Transformations Change the Analytical Scale

log1p(x)

Compresses a long right tail

Differences become multiplicative on the original scale

Standardisation

Centres and scales by mean and SD

Useful for optimisation; sensitive to extremes

Robust scaling

Uses median and IQR

More resistant to skew and extreme values

Binning

Converts a continuous variable to intervals

Can simplify communication but discards information

Winsorisation

Caps extreme values

Requires a defensible threshold and sensitivity check

Fit data-dependent transformations on training data and record their parameters

Subgroup Estimates Need Uncertainty

A large percentage difference in a small subgroup may be less reliable than a modest difference in a large subgroup.

Report together

- Subgroup sample size
- Estimate and confidence interval
- Missingness and coverage
- Relevant denominator

Check before interpreting

- Was the subgroup pre-specified?
- Are multiple comparisons being made?
- Could composition explain the difference?
- Will the subgroup exist at deployment?

Fairness analysis requires uncertainty, context and consequences, not only a table of group rates

Pandas Patterns for Reliable Analysis

Vectorise

Use arithmetic, masks and built-in methods

Faster and easier to audit than row loops

Method-chain

query → assign → groupby → agg

Makes transformations explicit and composable

Preserve types

category, nullable integers, datetime

Prevents silent coercion and reduces memory use

Validate

assertions and schema checks

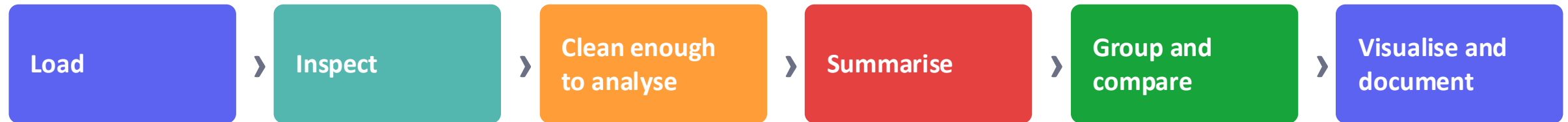
Fail early when assumptions no longer hold

Reproduce

fixed seeds, versioned inputs, run-all

Separates results from hidden notebook state

Treat an EDA notebook as an executable analytical record, not a sequence of manual experiments



Keep the workflow reproducible: rerun from the original source instead of relying on hidden notebook state.

Each code cell should answer a clear analytical question

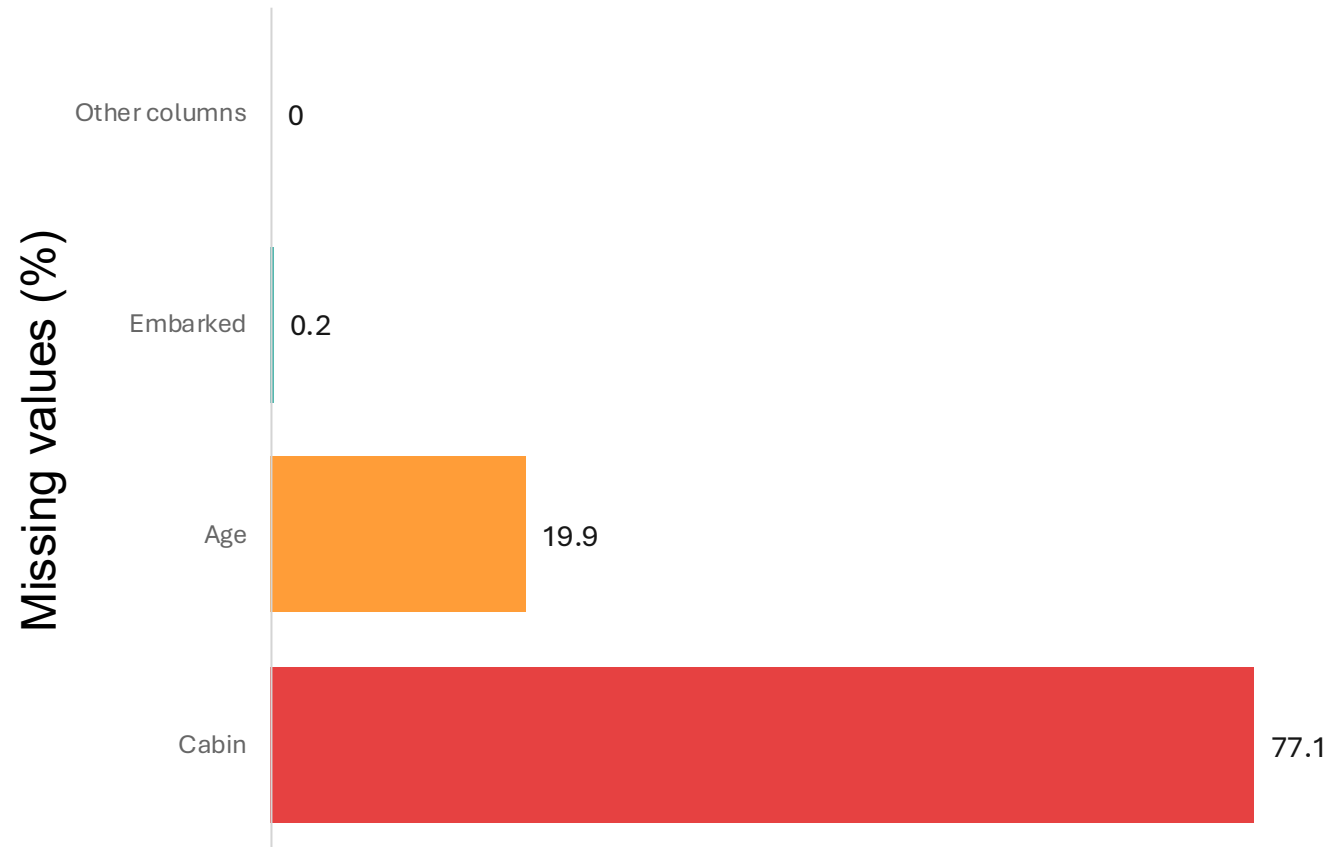
```
import pandas as pd

url = 'https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv'
df = pd.read_csv(url)

print(df.shape)
display(df.head())
display(df.dtypes)
df.info()
```

Check shape, column names, dtypes and sample records before calculating statistics

Profile Missing Values



```
missing = (  
    df.isna().mean()  
        .mul(100)  
        .sort_values(  
            ascending=False  
        )  
)
```

Missing percentages describe extent; group comparisons help interpret the pattern

```
# Select columns
view = df[['Survived', 'Pclass', 'Sex', 'Age', 'Fare']]

# Filter adults in first class
subset = df.loc[(df['Age'] >= 18) & (df['Pclass'] == 1)]

# Sort by fare
high_fares = df.sort_values('Fare', ascending=False).head(10)
```

Use .loc with explicit conditions so the analytical subset is visible and reproducible

```
summary = (  
    df.groupby(['Pclass', 'Sex'])  
        .agg(  
            passengers=('PassengerId', 'count'),  
            survival_rate=('Survived', 'mean'),  
            median_age=('Age', 'median'),  
            median_fare=('Fare', 'median')  
        )  
        .reset_index()  
)
```

groupby

defines comparable groups

agg

calculates named summaries

The grouping structure should follow the analytical question

Cross Tabs and Pivot Tables

Cross tab

- Counts or proportions for categorical variables
- Useful for outcomes across groups

Pivot table

- Aggregates a numerical measure across categories
- Supports multiple rows and columns

```
survival_table = pd.crosstab(df['Pclass'], df['Survived'], normalize='index')

fare_table = pd.pivot_table(
    df, values='Fare', index='Pclass', columns='Sex', aggfunc='median'
)
```

Label the aggregation and denominator so readers can interpret the table correctly

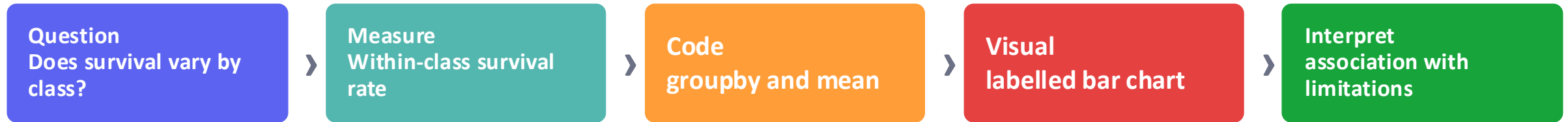
```
import matplotlib.pyplot as plt

ax = (df.groupby('Pclass')['Survived']
      .mean()
      .mul(100)
      .plot(kind='bar', color=['#54B7AF', '#FF9D38', '#E64141']))

ax.set_ylabel('Survival rate (%)')
ax.set_xlabel('Passenger class')
ax.set_title('Observed survival rate by passenger class')
plt.tight_layout()
```

A plot needs a question, readable labels, units and a cautious interpretation

Mini Analysis: From Question to Evidence



Evidence

- Observed rates differ substantially
- First class has the highest recorded rate

Limitations

- Class groups differ in age and sex composition
- Retrospective data does not identify a causal effect

A complete EDA statement combines calculation, visual evidence and limitations

Common Statistical and Pandas Mistakes

- Calculating a mean for an identifier or unordered category.
- Reporting a percentage without stating its denominator.
- Dropping missing rows without measuring who or what is removed.
- Using the full dataset to make preprocessing decisions before splitting.
- Interpreting correlation or a group difference as causal evidence.
- Producing many plots without linking them to analytical questions.

Correct code can still support an incorrect statistical conclusion

Summary and Preparation for Week 5

- Define the population, sample, observation and variable types before analysis.
- Use centre and spread together, choosing resistant summaries for skewed data.
- Summarise categories with clearly defined counts, proportions and denominators.
- Treat associations as descriptive evidence unless the design supports causal inference.
- Use pandas to build a transparent path from question to calculation, plot and interpretation.

Next week: deeper EDA, feature relationships and preparation for machine learning